

Integration of reciprocal functions



1 $f(x) = \ln x + C, x > 0$

2

a $-2 \ln x + C, x > 0$

b $6 \ln x + C, x > 0$

c $\frac{1}{7} \ln x + C, x > 0$

d $\ln(x + 5) + C, x > -5$

3

a $4 \ln x + C, x > 0$

b $-7 \ln x + C, x > 0$

c $\frac{3}{4} \ln x + C, x > 0$

d $\ln(x - 4) + C, x > 4$

e $2 \ln(x + 3) + C, x > -3$

f $\ln(2x + 3) + C, x > \frac{-3}{2}$

g $-2 \ln(x - 4) + C, x > 4$

h $\frac{1}{2} \ln(2x + 3) + C, x > -\frac{3}{2}$

i $-\ln(5 - x) + C, x < 5$

j $\frac{2}{3} \ln(3x + 5) + C, x > -\frac{5}{3}$

k $\ln(x^2 - 5) + C, x < -\sqrt{5}, x > \sqrt{5}$

l $8 \ln(x) + 5x + C, x > 0$

4

a $3 \ln 3$

b $\ln 2$

c $2 \ln \frac{7}{5}$

d $\frac{4}{3} \ln \frac{5}{2}$

e $5 \ln 2$

f $8 + \ln \frac{5}{3}$

g $\frac{1}{4} \ln \frac{17}{5}$

h $32 + \ln 2$

5

a $x^2 + x - 3 \ln x + C, x > 0$

b $5x + 2 \ln x + C, x > 0$

6

a $f'(x) = 1 + \ln x$

b $x \ln x - x + C, x > 0$

Reverse chain rule

7

a $f'(x) = \frac{2x}{x^2 - 4}$

b $3 \ln(x^2 - 4) + C$

8

a $f'(x) = \frac{2x + 2}{x^2 + 2x}$

b $\frac{1}{4} \ln(2x^2 + 4x) + C$

9

a $f'(x) = \frac{9x^2 + 10x}{3x^3 + 5x^2}$



b $2 \ln(3x^3 + 5x^2) + C$

10

a $\ln(4x - 3) + C$

b $5 \ln(3x - 2) + C$

c $-4 \ln(2x + 1) + C$

d $2 \ln(x^2 + 1) + C$

e $\ln(x^3 + 1) + C$

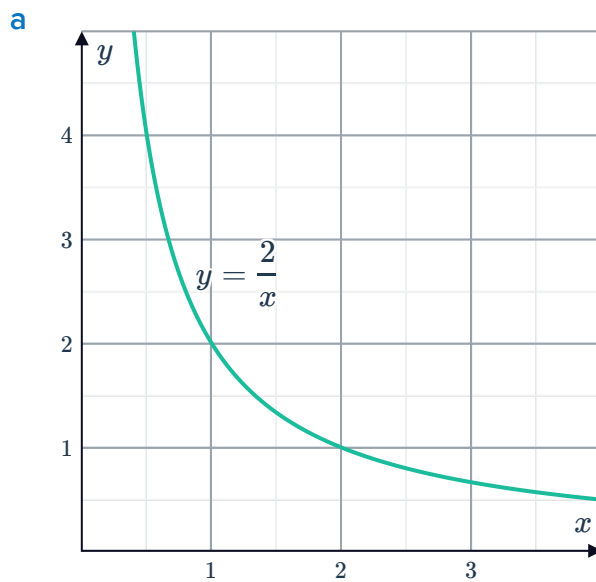
f $2 \ln(2x^2 - 3) + C$

g $\frac{1}{3} \ln(2x^3 + 7) + C$

h $\ln(2x^3 + 3x) + C$

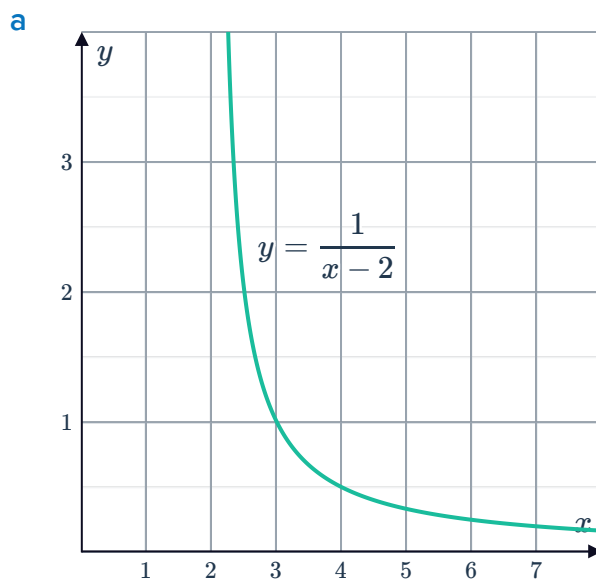
Area under the curve

11



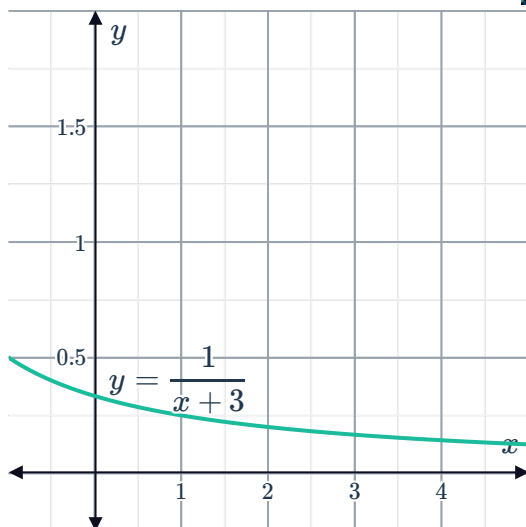
b $2 \ln \frac{3}{2} \text{ units}^2$

12



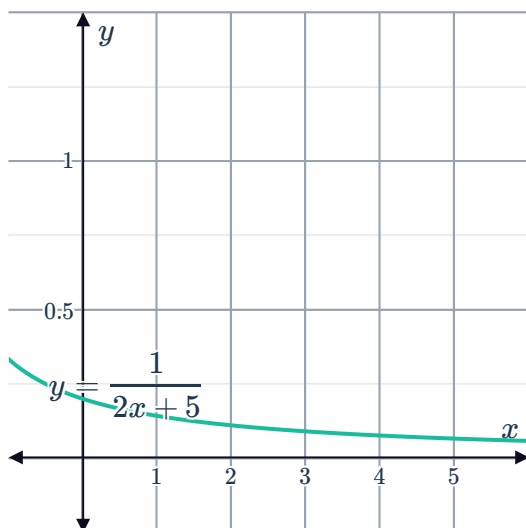
b $\ln 3 \text{ units}^2$

13 a



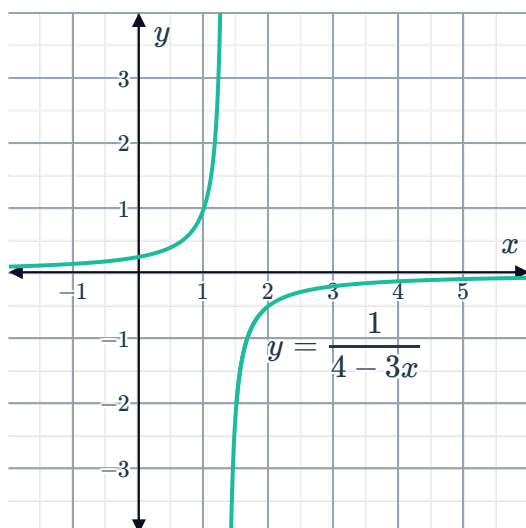
b $\ln \frac{7}{5}$ units²

14 a



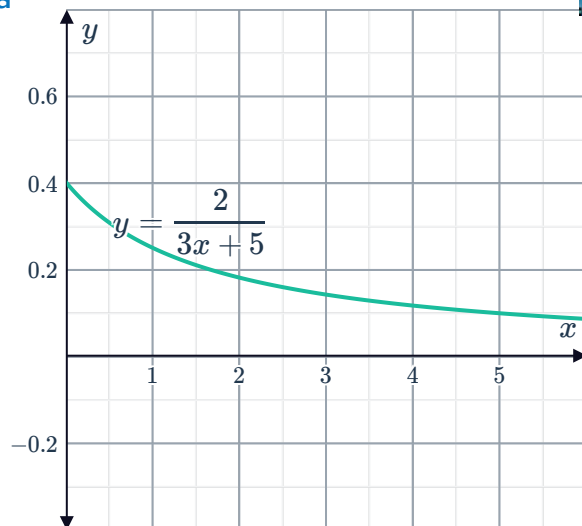
b $\frac{1}{2} \ln \frac{13}{11}$ units²

15 a



b $\frac{1}{3} \ln \frac{11}{5}$ units²

16 a



b $\frac{2}{3} \ln \frac{17}{5} \text{ units}^2$

17

a $(-\infty, 0) \cup (0, \infty)$

b $y \rightarrow \infty$

c $(70 + \ln 6) \text{ units}^2$

18 $2 + 5 \ln \frac{7}{5} \text{ units}^2$

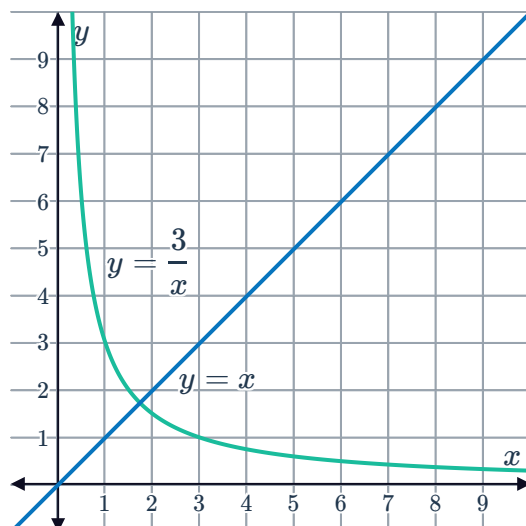
19

a $\frac{3x^4}{4} - 2x + \frac{1}{2} \ln x + C$

b $\frac{37}{4} + \frac{\ln 2}{2} \text{ units}^2$

20

a

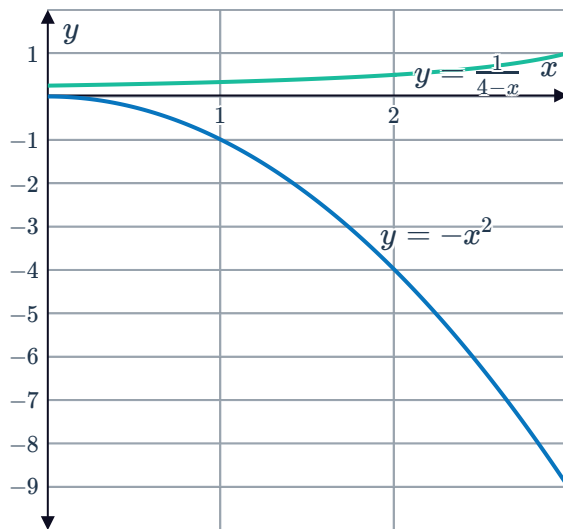


b $x = \sqrt{3}$

c $\frac{3}{2} + 3 \ln \sqrt{3} \text{ units}^2$

21

a

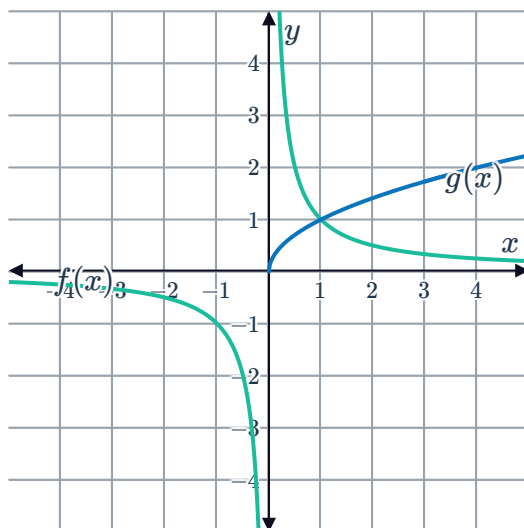


b

$$\left(\frac{26}{3} + \ln 3\right) \text{ units}^2$$

22

a

b $x = 1$

$$\text{c } \ln 2 + \frac{1}{3} \text{ units}^2$$

$$23 \quad k = e^{\frac{1}{3}}$$

24

$$\text{a } f'(x) = \frac{2x}{x^2 + 1}$$

$$\text{b } \frac{1}{2} \ln(x^2 + 1) + C$$

$$\text{c } \frac{1}{2} \ln 2 \text{ units}^2$$

25

$$\text{a } f'(x) = \frac{\cos x}{\sin x + 1}$$

$$\text{b } \ln(\sin x + 1) + C$$

$$\text{c } \ln\left(\frac{3}{2}\right) \text{ units}^2$$